

ADITYA CHAWLA

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EDUCATION

M.S. in Artificial Intelligence

San Jose State University, San Jose, CA, USA

Aug 2024 – May 2026

GPA: 3.85/4.0

Awards: Recipient of **Qualcomm Engineering Scholarship**, awarded by the College of Engineering.

Relevant Coursework: Natural Language Processing, Machine Learning on Graphs, , Deep Learning, Reinforcement Learning

B.E. in Electronics and Computer Engineering

Thapar Institute of Engineering and Technology, Patiala, India

Aug 2019 – June 2023

CGPA: 8.4/10.0

EXPERIENCE

Graduate Assistant at San Jose State University, San Jose

March 2025 - May 2026

- Drove the Office of the Registrar chatbots accessibility and user experience by translated SME insights from high-volume student queries (~3K–10K/month) into structured conversational flows and fallback ticketing.
- Skills:** Data Analysis, Data Visualization, User Data Management, PeopleSoft Campus Solutions, Process Optimization.

Software Engineering Associate at Amdocs, Gurugram

July 2023 – June 2024

- Leveraged **Generative AI and machine learning frameworks** to address a bottleneck in the Quotation system, resulting in a performance improvement of 40% as part of the invoicing team for the Amdocs Real-Time Billing application.
- Transitioned **the legacy codebase to C++11 standard**, which improved performance by 20% and reduced lines of code
- Owned the data engineering pipeline using PL/SQL scripts in oracle SQL developer for complex database operations and created custom UNIX scripts to automate secure retrieval of PII data from end users for Amdocs proprietary application.
- Skills:** C++, SQL, DBMS, UNIX, Shell Scripting, DevOps, Hadoop, PySpark, Data Pipelines, ETL, Performance Optimization

Software Developer Intern at 3rditech, IIT Delhi

Jan 2023 – June 2023

- Delivered **machine vision application** with the EBUS Software Development Kit, Microsoft Foundation Class (MFC) and C++ to optimize data delivery between host applications and vision standard-compliant imaging devices.
- Implemented **image acquisition, processing, and analysis algorithms** like CLAHE to extract meaningful information from high-performance imaging devices such as GigE Vision, USB3 Vision, and GenICam.
- Skills:** EBUS SDK, C/C++, Tkinter, Pillow, Pandas, Numpy, Keras, Image Processing, Algorithms, Computer Vision, Python.

PROJECTS

AgenticBI — AI-Powered BI Copilot Platform

Jan 2026 – Present

- Architected a BI-tool-agnostic event-driven AI **copilot** enabling multimodal (voice and text) interaction with business data, automating end-to-end BI workflows — query generation, execution, dashboarding, and insight explanation.
- Using a multi-agent **LangGraph** microservice system with self-correcting **LLM feedback loops**, **MCP** server for API-based tool orchestration, **GraphRAG** retrieval for schema-aware context grounding, and a Whisper-powered voice-input pipeline. Designed layered **guardrails** for prompt injection defence and secure SQL execution.
- Tech Stack:** SLM Agents, llama-3.3, Qwen-32B, Groq, MCP, A2A, ChromaDB, Vizro, LangChain, LangGraph, Docker, Git.

LLM Advisor – Trust-Based Model Recommender for Open-Source LLMs

Dec 2025 - Jan 2026

- Built a two-stage parallel evaluation and scoring pipeline that profiles **open-source LLMs**: Llama-3 8B, Qwen 2.5 7B, Kimi K2 on toxicity, bias, truthfulness, and safety to generate reproducible trustworthiness profiles for **ethical AI** deployment.
- Developed an **LLM-powered advisor** using Mixtral-8x7B that ingests model profiles and user priorities to recommend suitable LLMs, explaining trade-offs between safety, fairness, and factual reliability for real-world deployments.
- Tech Stack:** Groq-hosted LLMs, HuggingFace Datasets, Detoxify, sentence-transformers, Streamlit, Plotly, Python

Hierarchical Reinforcement Learning for Portfolio Management

Aug 2025 – Dec 2025

- Developed a **three-tier Hierarchical RL** system for multi-asset portfolio optimization, integrating market data and Reddit sentiment to achieve stronger risk-adjusted returns and reduced drawdowns across 3 years of back testing.
- Trained and aggregated **SAC, PPO, DDPG, and TD3** agents on **FinBERT** sentiment and Yahoo Finance pipelines, combining them through meta- and super-agent layers to outperform equal-weighted and single-agent baselines.
- Tech Stack:** PyTorch, gymnasium, stable-baselines3, transformers, FinBERT, yFinance, Reddit API, Python.